DP Biology – C4.2

Transfers of energy and matter

- Why can matter be recycled in ecosystems when energy cannot?
- What makes up for the energy that each group of living things in an ecosystem loses?

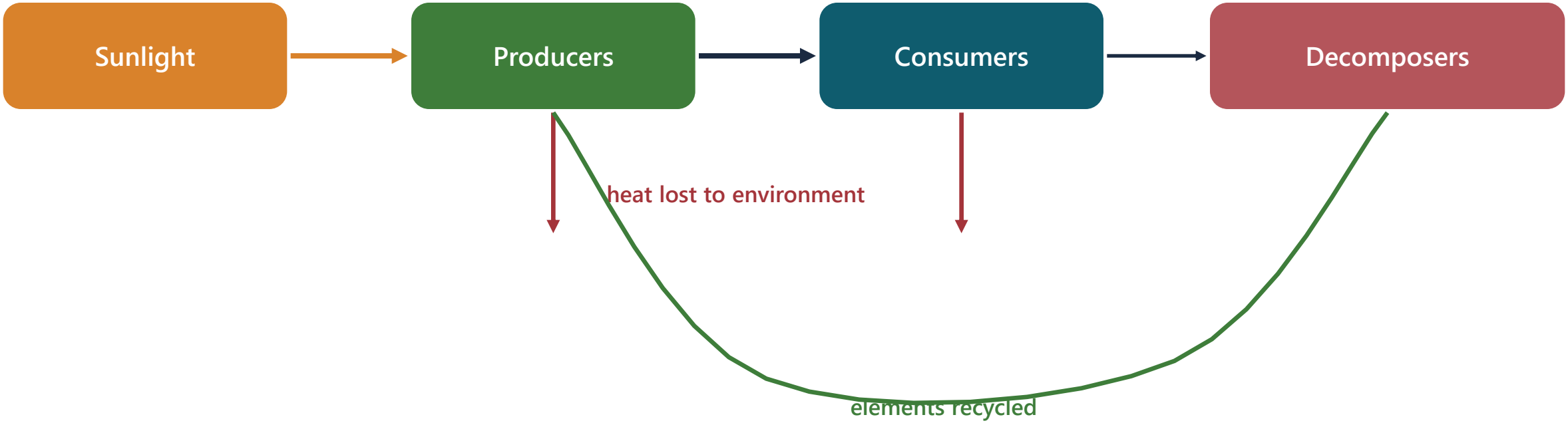
SL statements C4.2.1–22. Textbook pp. 578–601.
Image: tropical forest in Thailand. Vyacheslav Argenberg, CC BY 4.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	C4.2.1–10	Explain open systems and sunlight, draw food chains and webs, and compare autotrophs, heterotrophs and decomposers with trophic levels
2	C4.2.11–16	Draw energy pyramids, explain energy losses and short food chains, and compare primary and secondary production
3	C4.2.17–22	Draw the carbon cycle, explain sinks, sources and combustion, analyse the Keeling curve, and explain the recycling of elements

The big picture: energy flows, matter cycles



Big question

Why does every ecosystem need a constant new supply of energy, but not of carbon?

Ecosystems are open systems

C4.2.1

System	Energy	Matter
Open	In and out	In and out
Closed	In and out	Cannot enter or leave

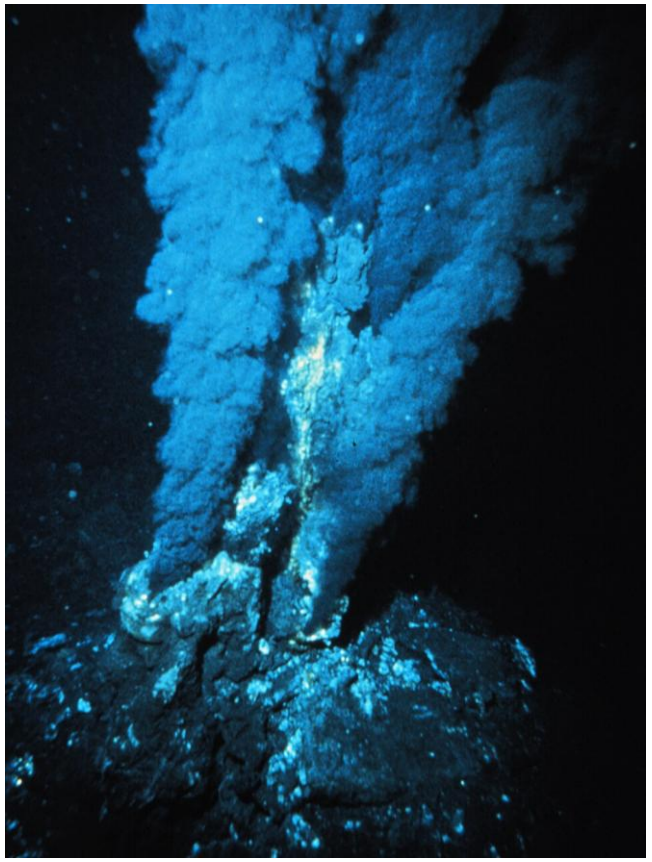
Ecosystems are **open**: sunlight in, heat out, animals and seeds moving. A sealed bottle garden is nearly **closed**.

Discuss

A sealed bottle garden has lasted decades. What must enter and leave? Why can it survive with no new matter?

Sunlight sustains most ecosystems

C4.2.2



Black smoker vent. P. Rona, OAR/NURP, public domain.

Producers capture **light**; consumers depend on it indirectly.

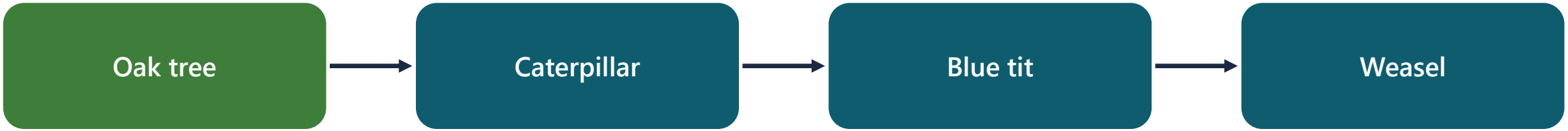
Exceptions

Caves get dead matter brought in by streams and bats. At deep-sea vents, bacteria oxidize sulfur compounds to make carbon compounds that support whole communities.

In water, light is absorbed with depth, so producers live in the upper layers.

Food chains

C4.2.3



A food chain shows the flow of **chemical energy**: each organism feeds on the previous one.

Common mistake

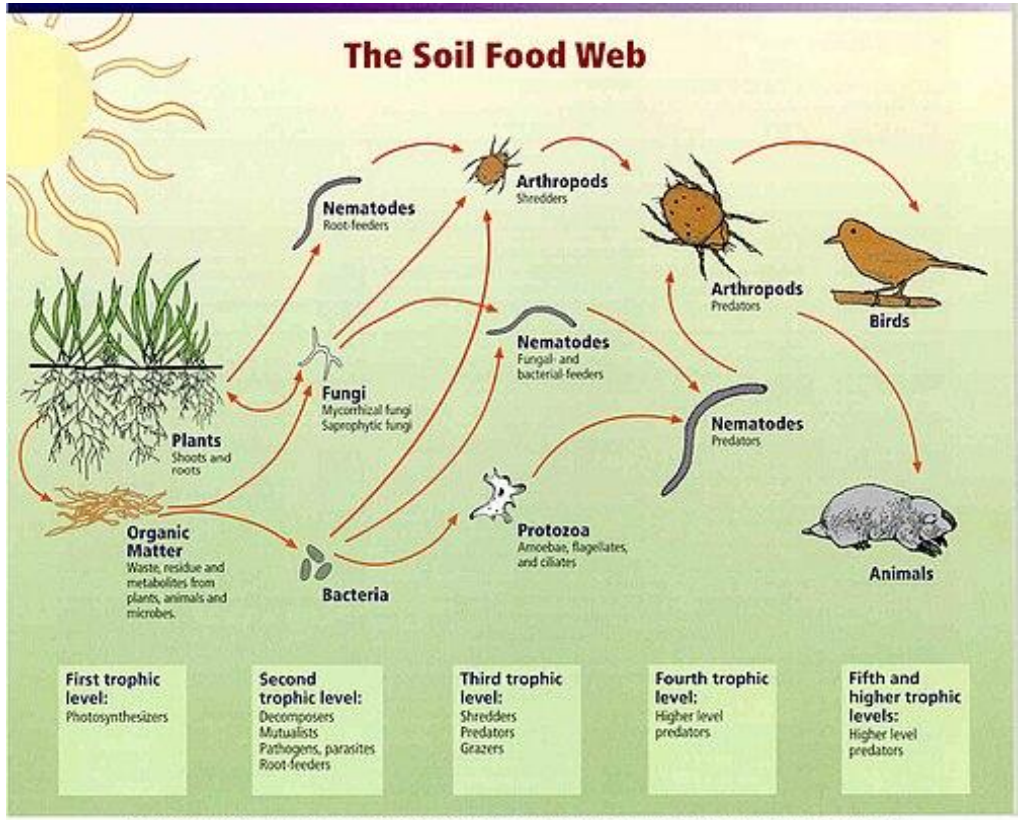
Arrows point to the organism that eats: the direction energy moves.

Activity

Build a local food chain with at least four organisms.

Food webs

C4.2.4



Relationships between soil food web, plants, organic matter, and birds and mammals
Image courtesy of USDA Natural Resources Conservation Service
http://soils.usda.gov/sqi/soil_quality/soil_biology/soil_food_web.html.

The soil food web. USDA, public domain.

A **food web** joins chains. Many consumers eat varied diets, so they appear in several places.

Activity

What might happen if one species were removed from a web?

Decomposers get energy from dead matter

C4.2.5



Earthworm. USDA NRCS, public domain.

Energy reaches decomposers in **dead organic matter**: faeces, dead parts, dead organisms.

Saprotrophs digest externally; detritus feeders ingest.



Fungi. Bernard Spragg, public domain.

Key idea

Decomposers respire, releasing energy, and return elements to the environment.

Autotrophs

C4.2.6–7

Autotrophs use an external energy source to make carbon compounds from inorganic molecules.

Type	Energy source	Examples
Photoautotroph	Light	Plants, algae, cyanobacteria
Chemoautotroph	Oxidation reactions	Iron- and sulfur-oxidizing bacteria

Iron bacteria

Oxidizing Fe^{2+} to $Fe^{3}+$ releases energy to make ATP, which powers CO_2 fixation.

Heterotrophs, and energy release

C4.2.8–9

Heterotrophs take carbon compounds from other organisms, digest them and assimilate the products.

	Consumers	Saprotrophs
Food	Living or recently killed	Dead matter
Digestion	Mostly internal	External

All respire

Autotrophs and heterotrophs both release energy by oxidizing carbon compounds. Plants respire all the time.

Trophic levels

Level	Name	Example
1	Producer	Oak
2	Primary consumer	Caterpillar
3	Secondary consumer	Blue tit
4	Tertiary consumer	Weasel

Remember

Omnivores feed at more than one level.

Note

Decomposers feed on dead matter from every level, so they are not given a trophic level.

Check-in: class 1

C4.2.1–10

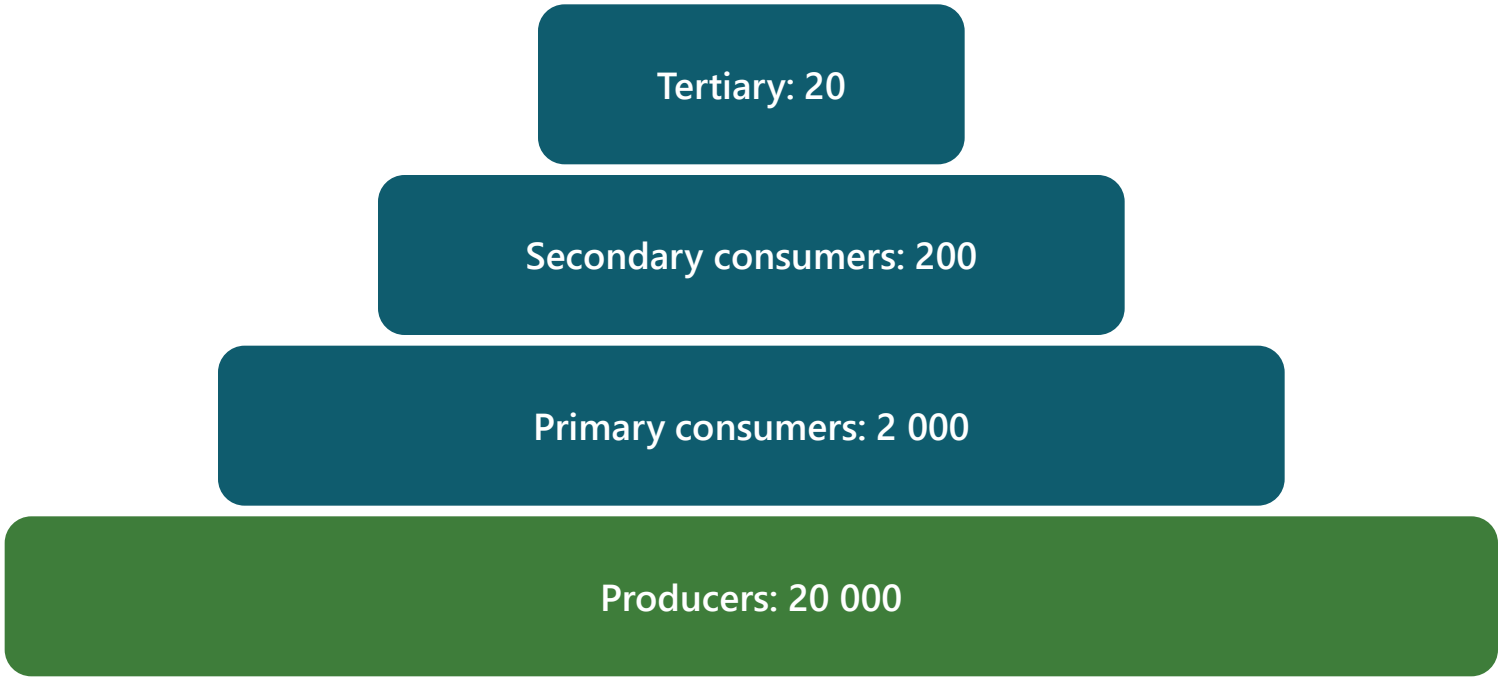
1. Distinguish open and closed systems.
2. Name the main energy source for ecosystems, and one exception.
3. Explain what arrows in a food chain show.
4. Distinguish photoautotrophs and chemoautotrophs.
5. Explain why decomposers have no trophic level.

How to use this

Answer without your notes, then check against slides 3–11.

Energy pyramids

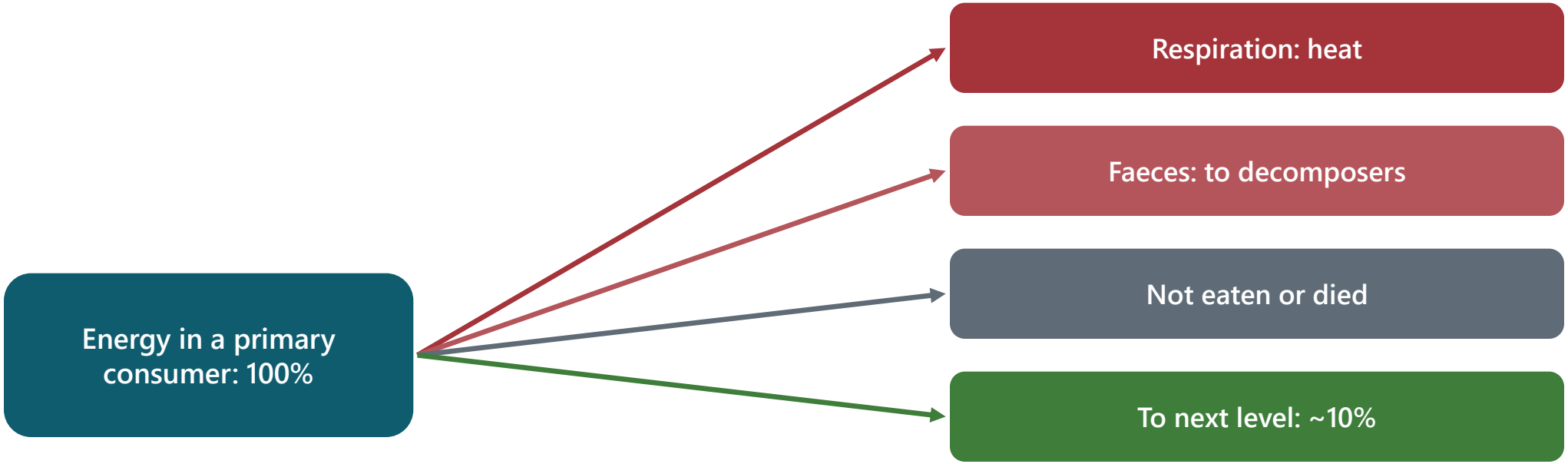
Energy per unit area per unit time, in kJ m^{-2} yr^{-1}. Invented values for practice.



- Stepped bars, not a triangle
- Producers at the bottom
- Label every level
- To scale where possible
- State the units

Where the energy goes

C4.2.12



Rule of thumb

About 10% passes on (typically 5–20%). Efficiency = energy at higher level / energy at lower level × 100.

Respiration turns chemical energy into heat

C4.2.13

No energy transfer is 100% efficient. Making and using ATP releases **heat**, which passes to the environment and cannot be turned back into chemical energy.

Big question

Matter is recycled because atoms are conserved and move between organisms and the environment. Energy is not: it enters as light and leaves as heat. New sunlight must keep arriving.

Why food chains are short

C4.2.14

So much energy is lost at each stage that too little remains for more levels. Each level has **less biomass**, not less energy per gram.

Common mistake

Higher levels are not poorer in energy per gram; there is just much less biomass.

Discuss

Why does land feed more people growing crops than raising cattle?

Primary and secondary production

C4.2.15–16

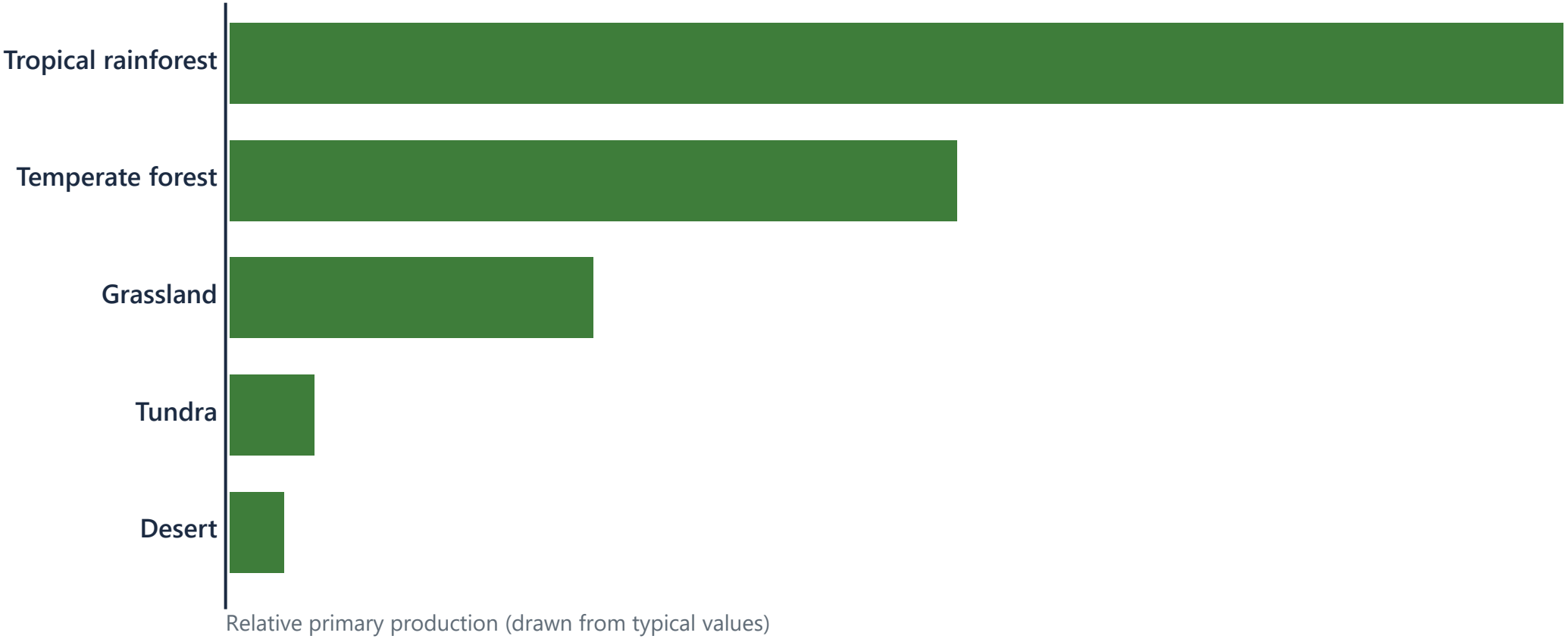
	Primary production	Secondary production
Organisms	Autotrophs	Heterotrophs
Carbon from	CO_2	Food
Amount	Higher	Lower
Units	g m^{-2} yr^{-1}	g m^{-2} yr^{-1}

Key idea

Secondary production is lower because carbon compounds are lost as CO_2 and water in respiration.

Primary production by biome

C4.2.15



Link

Climate decides biome (B4.1), so climate drives production.

Working with data: an energy pyramid

C4.2.11–12

Energy flow in a grassland, kJ m^{-2} yr^{-1}. Invented values for practice.

Trophic level	Energy
Producers	24 000
Primary consumers	2 600
Secondary consumers	230
Tertiary consumers	21

1. Efficiency, producers to primary?
2. Efficiency, primary to secondary?
3. Why can an energy pyramid never be inverted?

Check-in: class 2

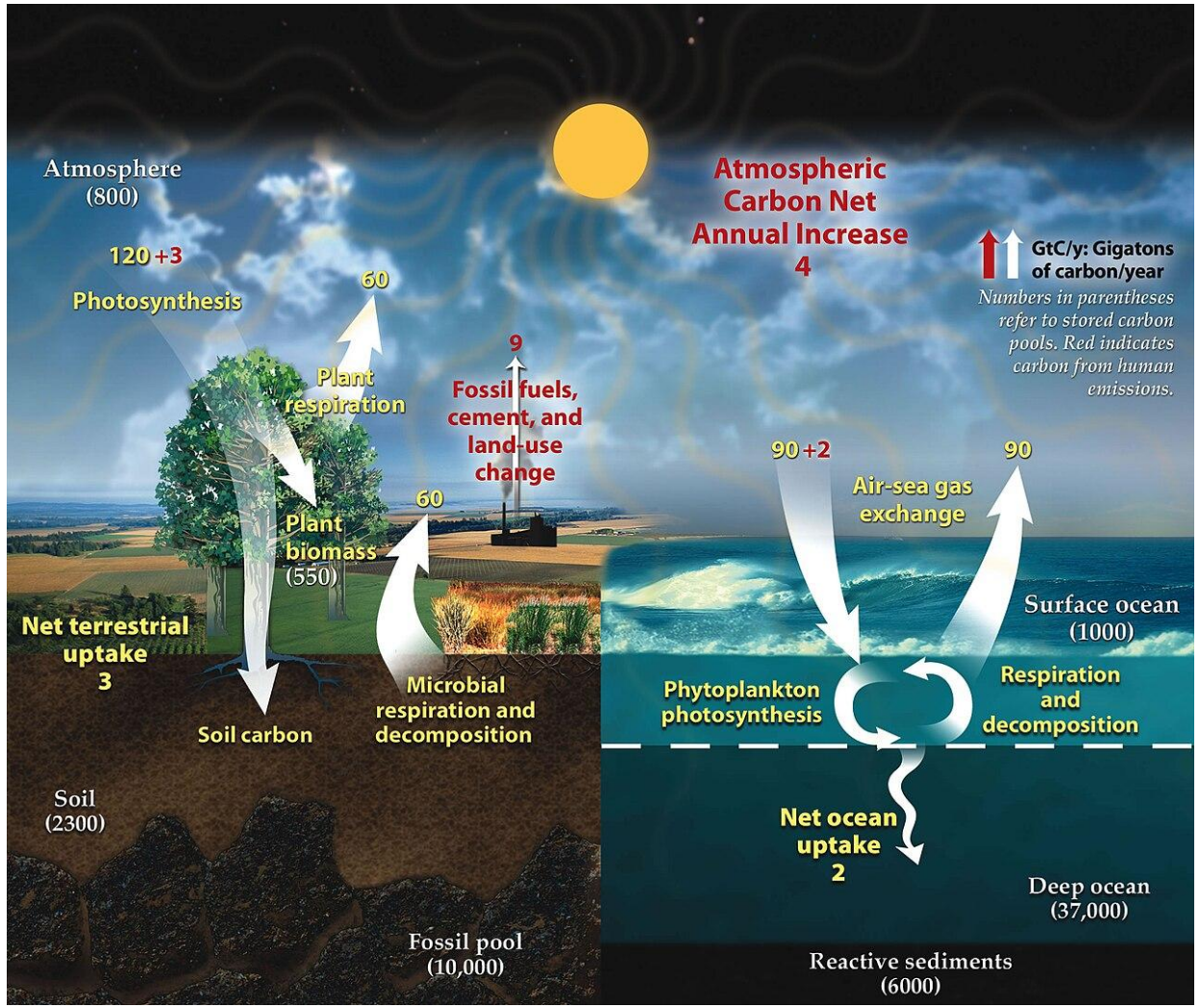
C4.2.11–16

1. Give the rules for drawing an energy pyramid.
2. Give three reasons energy is lost between levels.
3. Explain why energy cannot be recycled.
4. Explain why food chains are short.
5. Distinguish primary and secondary production.

How to use this

Answer without your notes, then check against slides 13–19.

The carbon cycle



The carbon cycle. Adapted from US DOE, public domain.

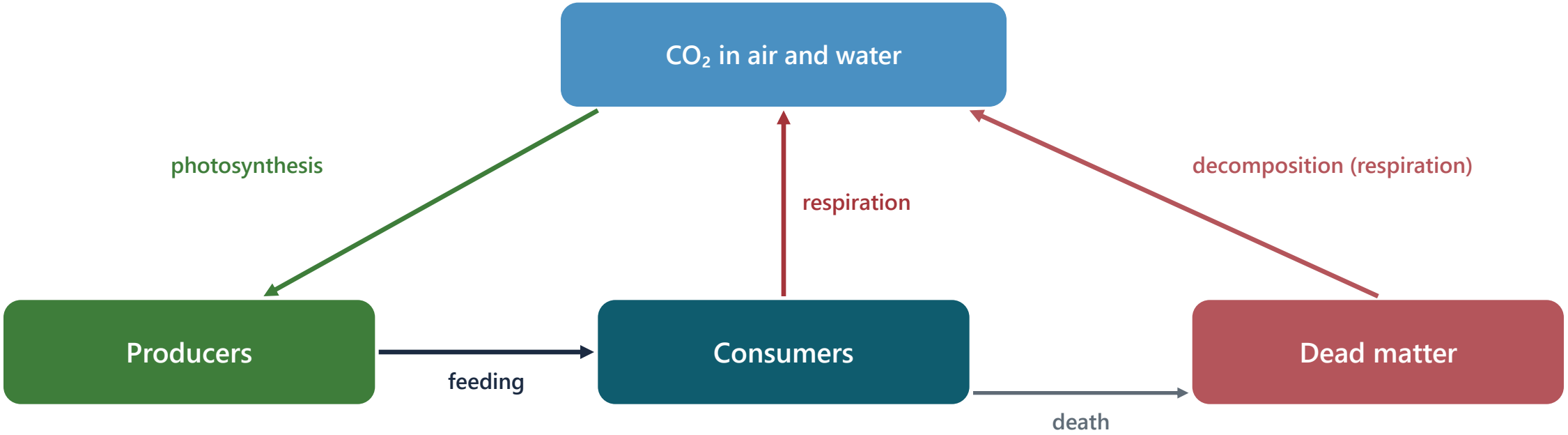
- CO_2 in air and water
- Photosynthesis
- Feeding
- Respiration
- Decomposition

Terms

A **pool** stores carbon; a **flux** is the rate it moves between pools.

Draw it yourself

C4.2.17



Exam tip

Label every arrow with a process; include respiration by producers too.

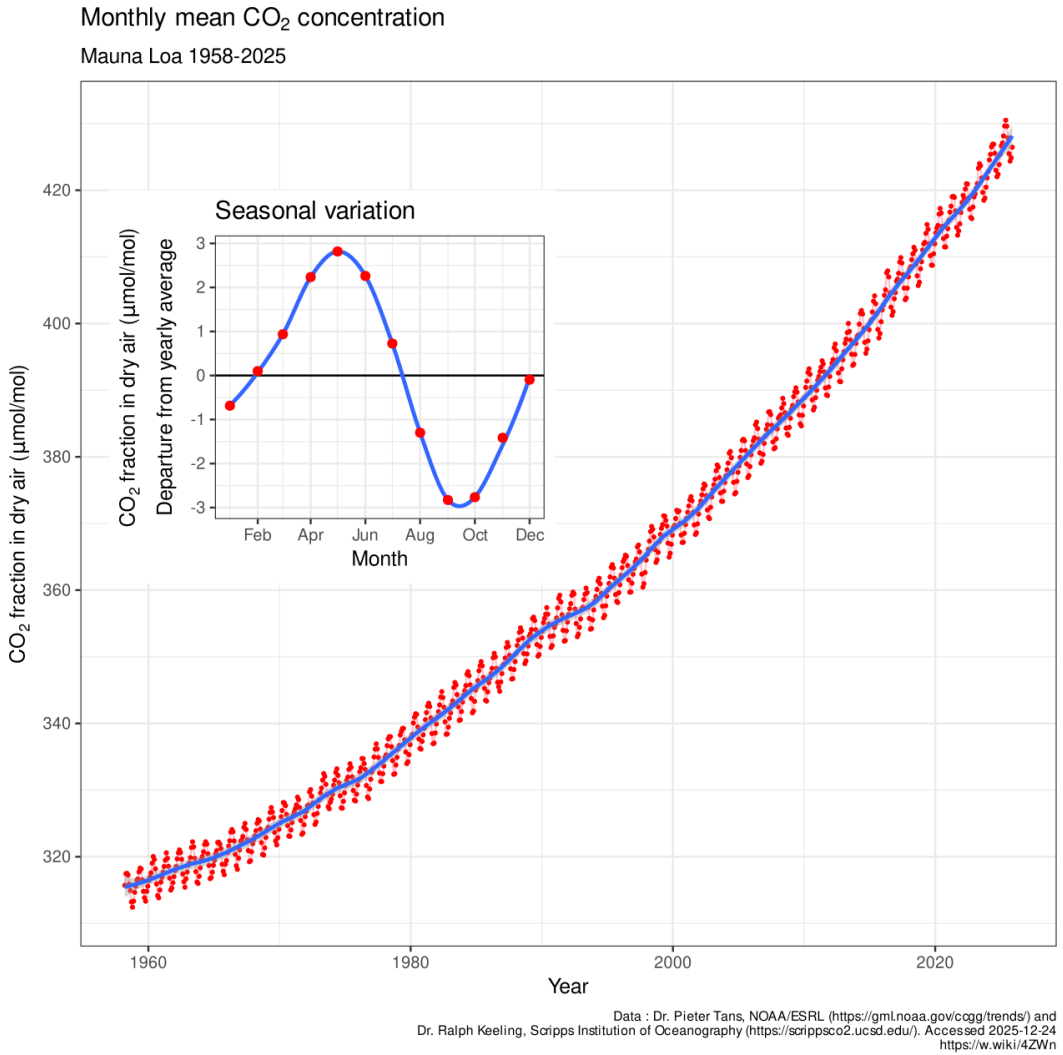
Sinks, sources and combustion

C4.2.18–19

Ecosystem is a	When	Net effect
Carbon sink	Photosynthesis > respiration	CO_2 taken up
Carbon source	Respiration > photosynthesis	CO_2 released

Fuel	Origin	Formed
Peat	Part-decayed plants in waterlogged ground	Thousands of years
Coal	Buried, compressed peat	Hundreds of millions of years
Oil and gas	Buried marine organisms	Millions of years

The Keeling curve



Carbon dioxide at Mauna Loa. Oeneis, CC BY-SA 4.0.

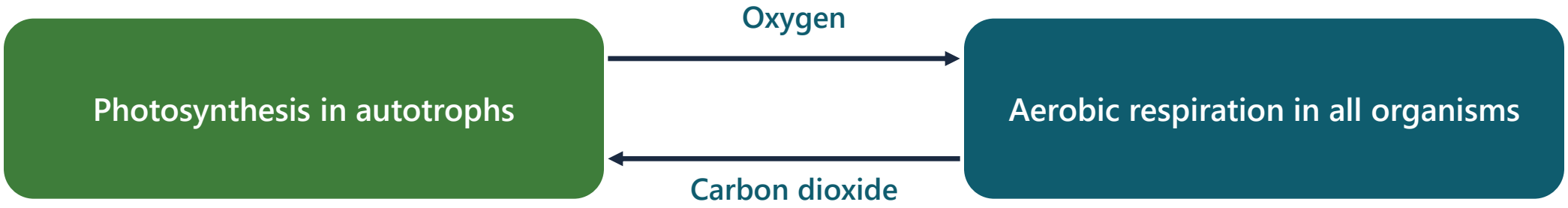
Feature	Explanation
Long-term rise	Fossil fuel combustion, deforestation
Yearly wave	Down in northern summer (photosynthesis); up in winter (respiration)

Why north?

Most land, and so most plants, is in the northern hemisphere.

Photosynthesis and respiration depend on each other

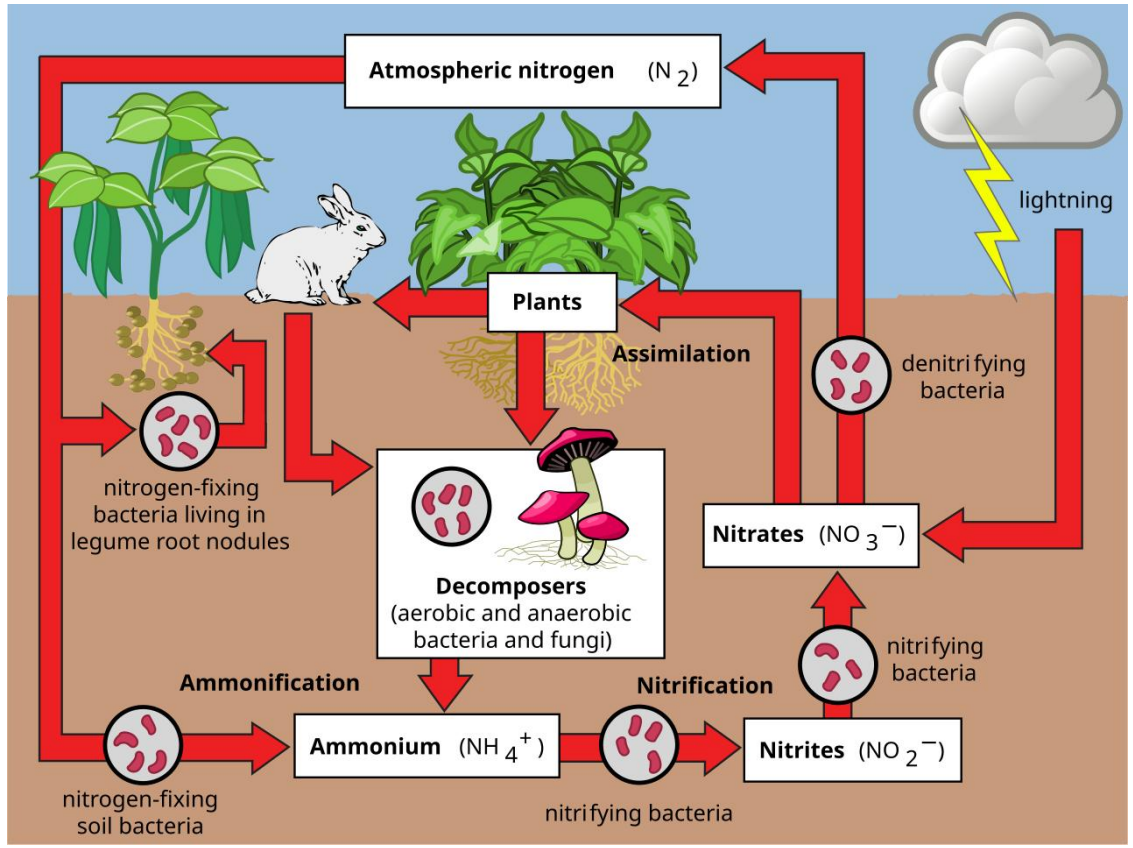
C4.2.21



Key idea

Before photosynthesis there was little oxygen. Oxygen-producing photosynthesis made aerobic respiration, and large active organisms, possible.

All elements are recycled



The nitrogen cycle. CC BY-SA 3.0.

Every element organisms need is recycled.

Autotrophs take elements from air, water and soil; heterotrophs from food; **decomposers** return them.

Key idea

Know the general principle, not every detail of each cycle.

Working with data: CO_2 at Mauna Loa

C4.2.20

Year	Mean CO_2 / ppm
1960	317
1980	339
2000	370
2020	414

1. Calculate the mean yearly rise 1960–2020.
2. Is the rise speeding up? Use the data.
3. Explain the cause.

Practice: sink or source?

C4.2.18

Classify

A young growing forest.

Classify

A drained, drying peat bog.

Explain

How a forest fire changes a sink.



Sphagnum moss in a bog. NC Wetlands, CC0.

Discuss

Why does waterlogged peat store carbon? What happens when it dries? (Link D4.3.)

Practice: carbon

C4.2.17–22

Draw

The carbon cycle with five processes.

Explain

The yearly wave on the Keeling curve.

Explain

Why decomposers matter for recycling.

Use

pool, flux, sink, source, combustion

Check-in: class 3

C4.2.17–22

1. Name five processes in the carbon cycle.
2. Distinguish a carbon sink and a carbon source.
3. Name three fossil fuels and their origins.
4. Explain both features of the Keeling curve.
5. Explain why decomposers are key to recycling.

How to use this

Answer without your notes, then check against slides 21–29.

Common mistakes to avoid

Not this	But this
Energy is recycled.	Energy flows; matter cycles.
Energy is destroyed.	It becomes heat, which leaves.
Arrows show who eats whom.	They show energy flow.
Plants only respire at night.	They respire all the time.

Summary: energy and matter

	Key idea
C4.2.1–5	Open systems; sunlight; chains, webs, decomposers
C4.2.6–10	Autotrophs, heterotrophs, trophic levels
C4.2.11–14	Pyramids; losses; heat; short chains
C4.2.15–16	Primary and secondary production
C4.2.17–19	Carbon cycle; sinks, sources; combustion
C4.2.20–22	Keeling curve; interdependence; recycling

Check that you can...

1. **Explain** energy flow through chains and webs
2. **Construct** energy pyramids
3. **Explain** energy losses
4. **Distinguish** primary and secondary production
5. **Describe** the carbon cycle
6. **Interpret** the Keeling curve

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why energy cannot be recycled in an ecosystem.
2. State three reasons for energy loss between trophic levels.
3. Explain why atmospheric carbon dioxide at Mauna Loa falls during the northern summer.

Image credits

Virgin tropical forest in southern Thailand, Khlong Phanom, Surat Thani.jpg – Vyacheslav Argenberg – CC BY 4.0

Blacksmoker in Atlantic Ocean.jpg – P. Rona / OAR/National Undersea Research Program (NURP); NOA – Public domain

Soil food webUSDA.jpg – USDA – Public domain

Earthworm in a No-till Field near Brookings, SD (14366341997).jpg – USDA NRCS South Dakota – Public domain

Mycena haematopus. (42029182981).jpg – Bernard Spragg. NZ from Christchurch, New Zealand – Public domain

Carbon cycle.jpg – Diagram adapted from U.S. DOE, Biological and Environmental – Public domain

Mauna Loa CO2 monthly mean concentration.svg – Oeneis – CC BY-SA 4.0

Nitrogen Cycle 2.svg – File:Nitrogen Cycle.svg: **Cicle_del_nitrogen_de.svg:** Cicle_ – CC BY-SA 3.0

Sphagnum moss in Panthertown Valley bog.jpg – NC Wetlands from Raleigh, NC – CC0

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